

Investigating the Impact of Market Shifts on Stock Performance of Leading NASDAQ Companies

Group Members: Bhuvan Dubey (30139765) and Akaljot Sangha (10106725)

Purpose

This project focuses on how major technology and consumer companies in the NASDAQ stock market have changed in performance over recent years. Stock prices constantly shift because of investor behavior, economic changes, and major world events, so looking at real stock market data helps us understand those trends.

This project is interesting because it helps investors, analysts, and policymakers understand how technology leaders respond to global shocks. Most depth analysis is used internally by firms only, so public information is usually very surface level.

Key Questions

We will investigate

- Which company experienced the largest change in its average annual return over the last 3–5 years?
 - This question aims to uncover which of the top 10 NASDAQ companies demonstrated the strongest growth or decline in returns when comparing early and recent years in the dataset
- How did each company's volatility (daily return variability) change before and after the onset of the COVID-19 pandemic?
 - This investigates how major market events, like COVID-19, influenced investor behavior and price stability in leading NASDAQ companies.

This analysis can help us better understand:

- Which companies are gaining (or losing) momentum
- How risky certain stocks became after COVID-19
- How real-world events influence investor confidence

This project also connects our SQL skills to a real financial dataset and real-world decision making.

Population of Interest

All available stock records in the dataset from these companies:

- Apple, Microsoft, Tesla, Amazon, Meta, Netflix, Starbucks, Cisco, Qualcomm, AMD

These are some of the most influential companies in the U.S. tech market.

Variables We Will Use

The main variables we will analyze from the dataset include the daily closing stock price, which tells us how much each company's stock was valued at the end of each trading day, and daily

return, which represents the percentage change in price from one day to the next. We will also use the date field to organize performance by year and separate the data into pre-COVID and post-COVID periods (March/2020 as the Covid start month). We will use +/- 3 years (2017->2023) as our data range. The company name column will help us compare performance across the top NASDAQ companies included in this project. Additionally, the trading volume field will be considered as a secondary measure of market activity and investor interest. All of these variables together will allow us to calculate trends in growth and volatility over time for each company.

How the Data Was Collected

The dataset contains daily stock information gathered directly from Nasdaq.com, and compiled by a Kaggle contributor. Each row in the dataset represents one trading day for one company.

Dataset Description

Data

- Dataset Name: Stock Market Historical Data of Top 10 Companies
- Website: Kaggle
<https://www.kaggle.com/datasets/khushipitroda/stock-market-historical-data-of-top-10-companies?resource=download>
- Format & Structure:
 - CSV file
 - Columns include: Company, Date, Close/Last, Volume, Open, High, Low
- Size: 1 file, approximately 1.4 MB

Accessibility

The dataset appears complete with minimal missing values, and no known licensing restrictions, making it suitable for reproducible analysis.

Time Frame

The data spans from 2013-2023, which allows us to split the data around the COVID-19 pandemic. We will only be using 2017-2023 for our analysis.

Topics We Will Investigate

3.1 Main Statistical Focus

Question 1: Change in average returns

We will compare earlier years in the dataset vs. the most recent years to see which company improved (or declined) the most.

Null Hypothesis: There is no significant change in annual returns over time.

Alternative Hypothesis: At least one company shows a significant change.

Question 2: Change in volatility

We will compare pre-COVID vs. post-COVID volatility for each company.

Null Hypothesis: Volatility did not significantly change after COVID-19 began.

Alternative Hypothesis: Volatility significantly increased or decreased.

Background and Context

The stock market is a valuable contributor to economic stability, particularly for large, well-known NASDAQ companies, including Apple, Microsoft, Tesla and Amazon. Over the last several years, these companies have undergone significant changes, as demand for technology has increased and investment behavior has changed. One of the most significant impacts on stock performance in recent years has been the COVID-19 pandemic that led to drastic declines during early 2020 and a sudden boom in volatility (Zhang et al., 2020). Many tech companies quickly recovered as reliance on digital services increased, including online shopping, digital media consumption, and virtual applications for work (Baker et al., 2020). By calculating average annual returns and average percentage volatility before and after the COVID-19 shock, we can determine which companies grew the most and how risky they've become during this time of uncertainty. This ultimately provides an enhanced perspective of how real-world events can affect investor sentiment and stock performance.

Statistical Methods

To answer our research questions, we plan to:

1. Import and clean the dataset in a SQL database
 - a. Convert prices to numeric, ensure dates are correctly formatted
2. Calculate daily returns using SQL
 - a. $(\text{Close price} - \text{Previous Day Close}) / \text{Previous Day Close}$
3. Compute average annual returns for each company
 - a. Compare earlier years vs. recent years
4. Measure volatility using STDDEV() in SQL
 - a. Compare values before and after March 2020
5. Export results and create visualizations
 - a. Line charts of annual returns
 - b. Bar charts of volatility change

We will know our project is successful if our results clearly show:

- Which company had the biggest return change
- How volatility shifted after COVID-19
- SQL queries and visuals support our conclusions

We will consider our analysis successful if we can identify statistically significant differences in returns or volatility between time periods.

References

- Baker, S. R., Bloom, N., Davis, S. J., & Terry, S. J. (2020). COVID-induced economic uncertainty. *NBER Working Paper No. 26983*. <https://doi.org/10.3386/w26983>
- Pitroda, K. (2023, July 18). Stock market: Historical data of top 10 companies. Kaggle. <https://www.kaggle.com/datasets/khushipitroda/stock-market-historical-data-of-top-10-companies?resource=download>
- Zhang, D., Hu, M., & Ji, Q. (2020). Financial markets under the global pandemic of COVID-19. *Finance Research Letters*, 36, 101528. <https://doi.org/10.1016/j.frl.2020.101528>